

17 JULY 2025

Quick

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M T

Network

COMPUTER NETWORK

ABBREVIATIONS / ACRONYMS

- 1) RJ-45 = Registered Jack-45
- 2) CAN = Campus Area Network
- 3) OSI = Open System Interconnected
- 4) ISO = International System Interconnected
- 5) LLC = Logic Linked Control
- 6) MAC = Medium Access Control
- 7) MBps = Mega Bytes per second (1 MBps = 8 Mbps)
- 8) Mbps = Mega bits per second (1 Byte = 8 bits $\Rightarrow$ 8 Mbps = 1 MBps)
- 9) URL = Uniform Resource Locator
- 10) URI = Uniform Resource Identifier
- 11) HTTP = Hyper Text Transfer Protocol
- 12) HTTPS = Hyper Text Transfer Protocol Secure
- 13) DNS = Domain Name Systems
- 14) SMTP = Simple Mail Transfer Protocol
- 15) ~~SNTP~~ = SNMP = Simple Network Management Protocol
- 16) POP = Post Office Protocol
- 17) TELNET = No abbreviation
- 18) FTP = File Transfer Protocol
- 19) TFTP = Trivial File Transfer Protocol
- 20) TCP = Transmission Control Protocol
- 21) UDP = User Datagram Protocol
- 22) IP = Internet Protocol
- 23) ARP = Address Resolution Protocol
- 24) RARP = Reverse Address Resolution Protocol
- 25) ICMP = Internet Control Message Protocol
- 26) CSMA/CD = Carrier Sense Multiple Access / Collision Detection
- 27) PPP = Point to Point Protocol
- 28) SCTP = Stream Control Transfer Protocol
- 29) WLAN = Wireless LAN

- 30) PCI = Peripheral Component Interface
- 31) NIC = Network Interface Card
- 32) ~~Shielded~~ STP = Shielded Twisted Pair
- 34) DCE = Data Communication Equipment
- 35) DTE = Data Terminal Equipment
- 36) AP = Access Point
- 37) ESS = Extended Services Set
- 38) STA = Station
- 39) DS = Distributed System
- 40) BSS = Basic Service Set
- 41) BTS = Base Transceiver Station
- 42) MANET = Mobile Ad-hoc Network
- 43) VANET = Vehicular Ad-hoc Network
- 44) SDMA = Space Division Multiple Access
- 45) FDMA = Frequency Division Multiple Access
- 46) TDMA = Time Division Multiple Access
- 47) CDMA = Code Division Multiple Access
- 48) OFDM = Orthogonal Frequency Division Multiple
- 49) RF = Radio Frequency
- 50) BTS = Base Transceiver Station

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Network:

Collection of autonomous system

Types of Network:

- (i) PAN (Personal Area Network)
- (ii) LAN (Local Area Network)
- (iii) CAN (Campus Area Network)
- (iv) MAN (Metropolitan Area Network)
- (v) WAN (Wide Area Network)

Examples

PAN (Bluetooth)
LAN (Wifi)
CAN (colleges)

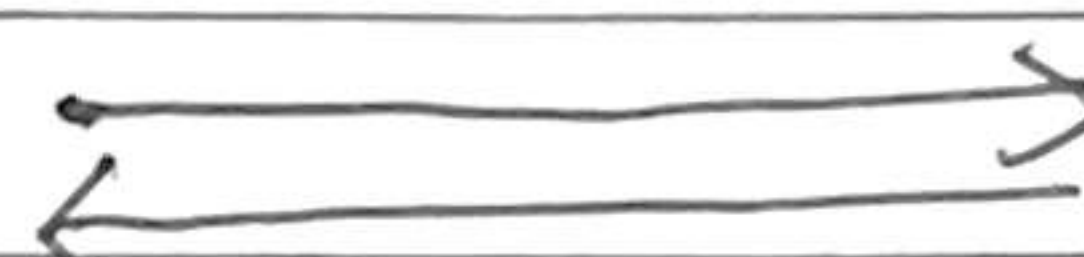
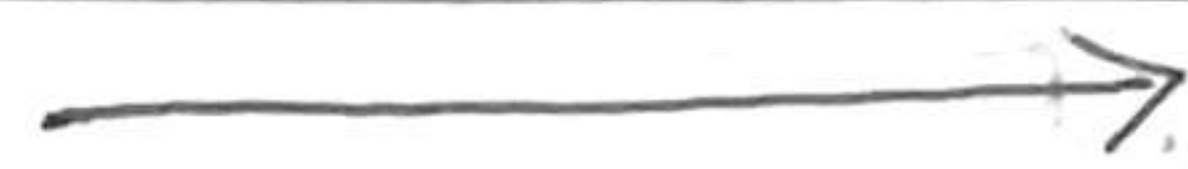
MAN (Smart cities)
WAN (Internet)

Range



Modes of Network:

- (i) Simplex (eg. Radio, Television)
- (ii) Half duplex (eg. Walkie-Talkie)
- (iii) Full duplex (eg. Mobile phones)



} Same frequency



Medium of Network:

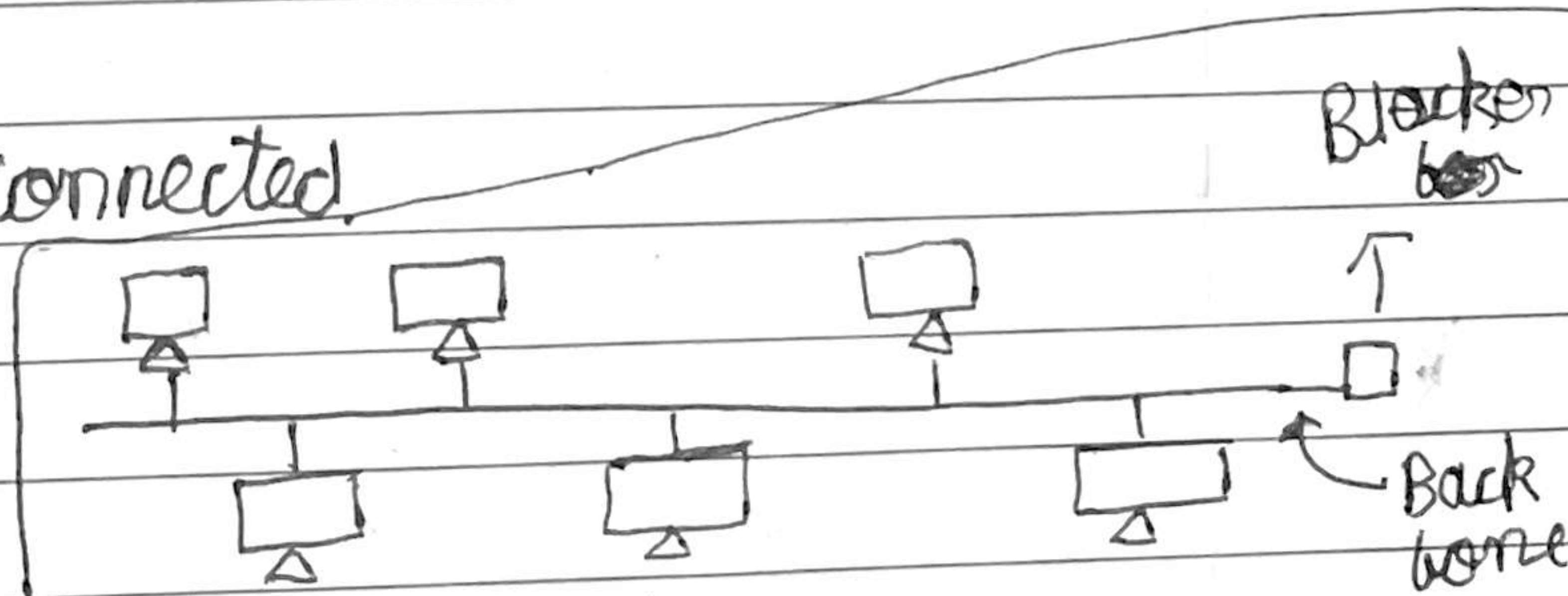
- (i) Wired
 - Coaxial (1-way)
- (ii) Wireless
 - Air
 - Twisted pair
 - Twisted Shielded
 - Unshielded
 - Optical fiber/fiber optics (light signals)

Shielded
4 pairs 8 wires RJ-45
Orange White/Orange Blue White/Blue Green White/Green Brown White/Brown

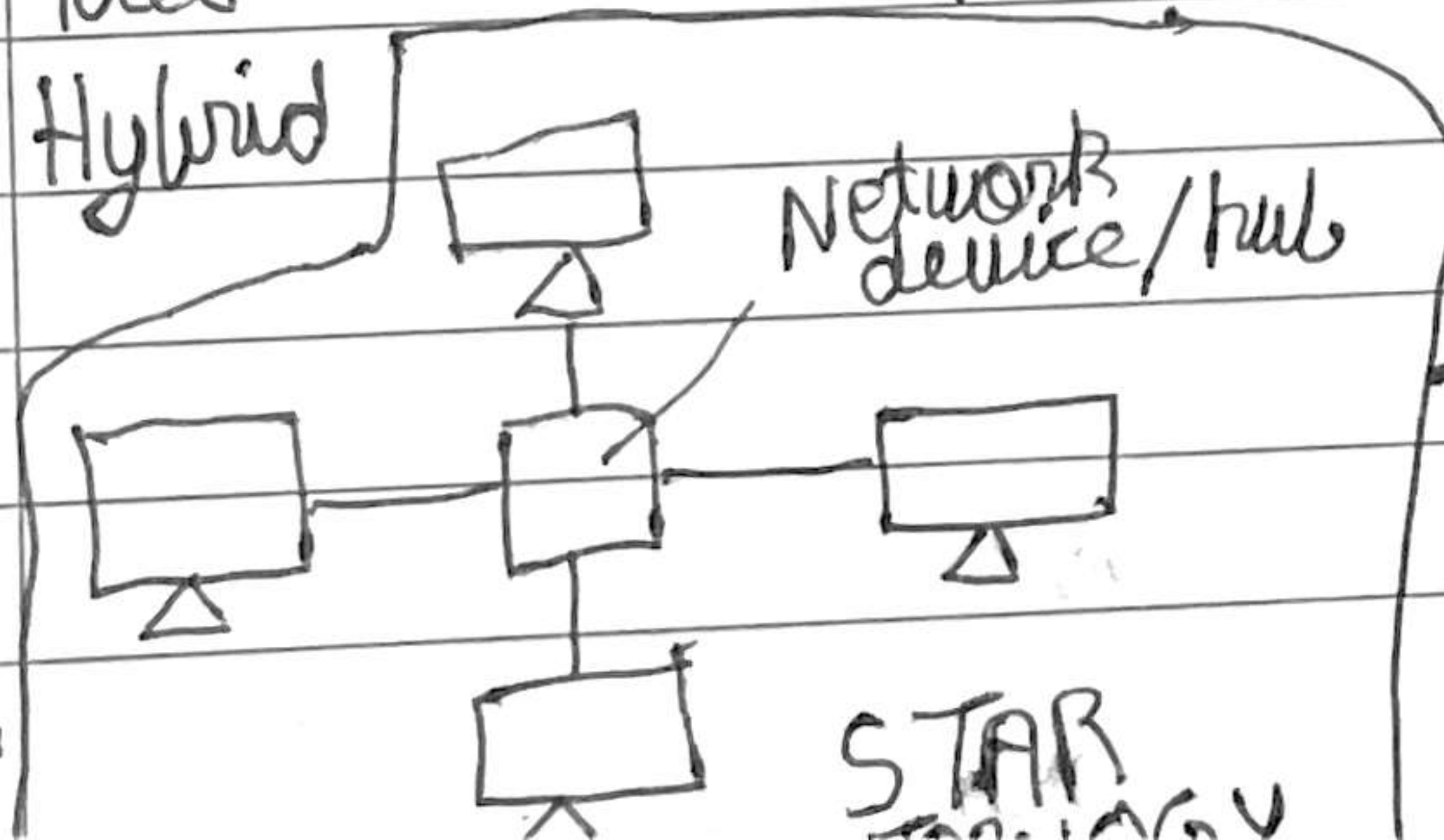
1 pair RJ11
2 wires

Topology:

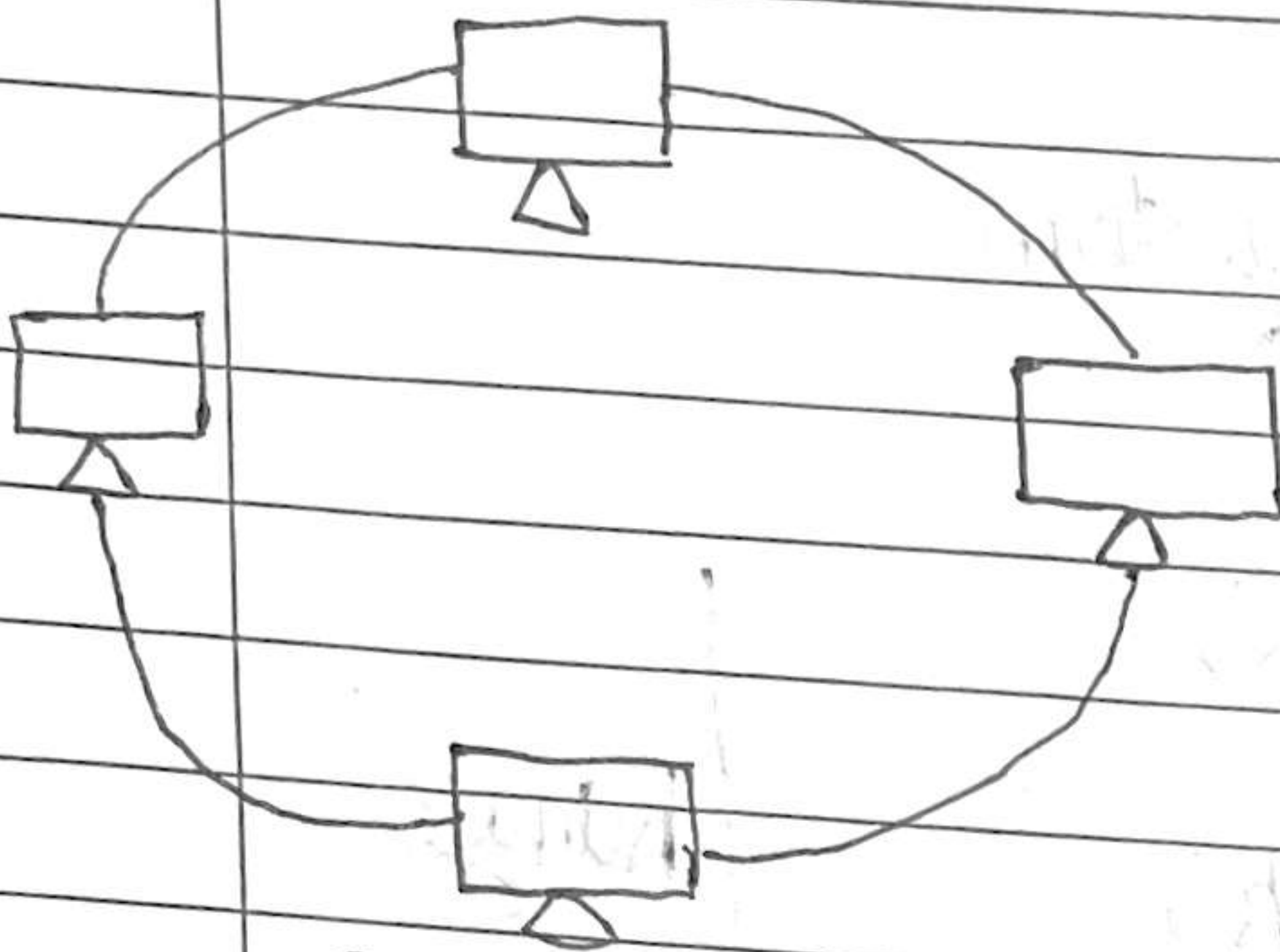
- (i) Bus
- (ii) Mesh / Strongly connected
- (iii) Star
- (iv) Ring
- (v) Tree
- (vi) Hybrid



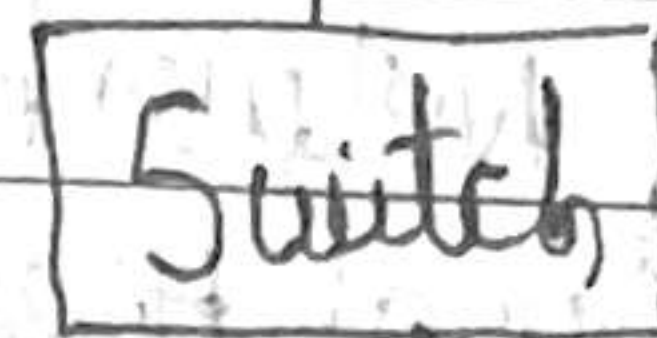
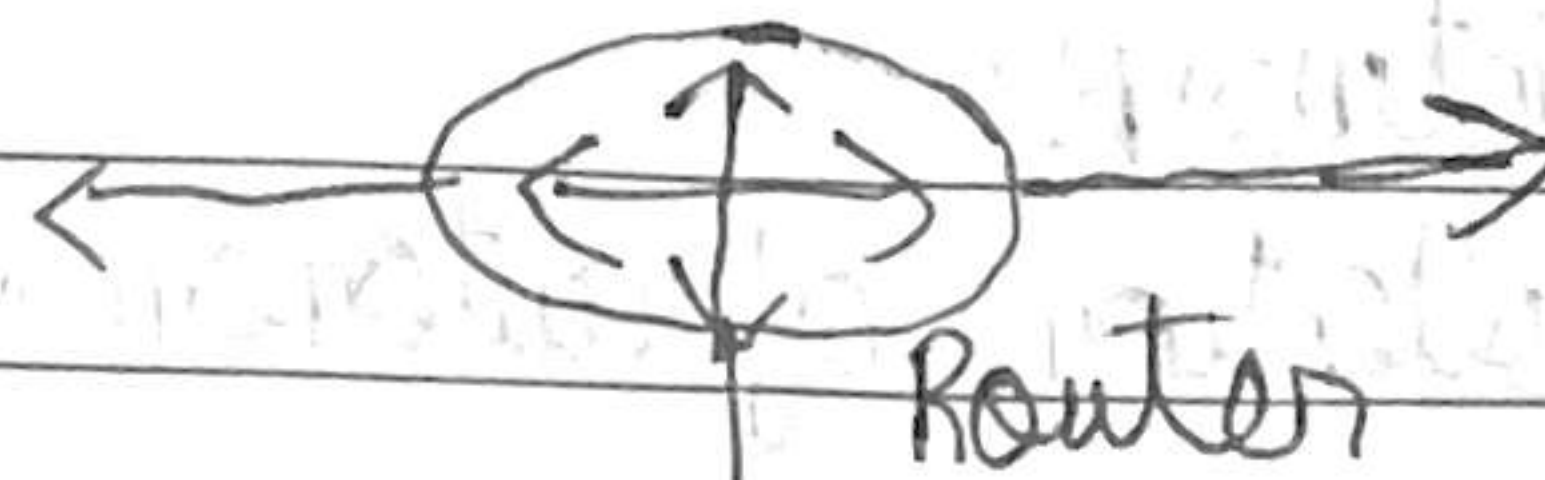
BUS TOPOLOGY



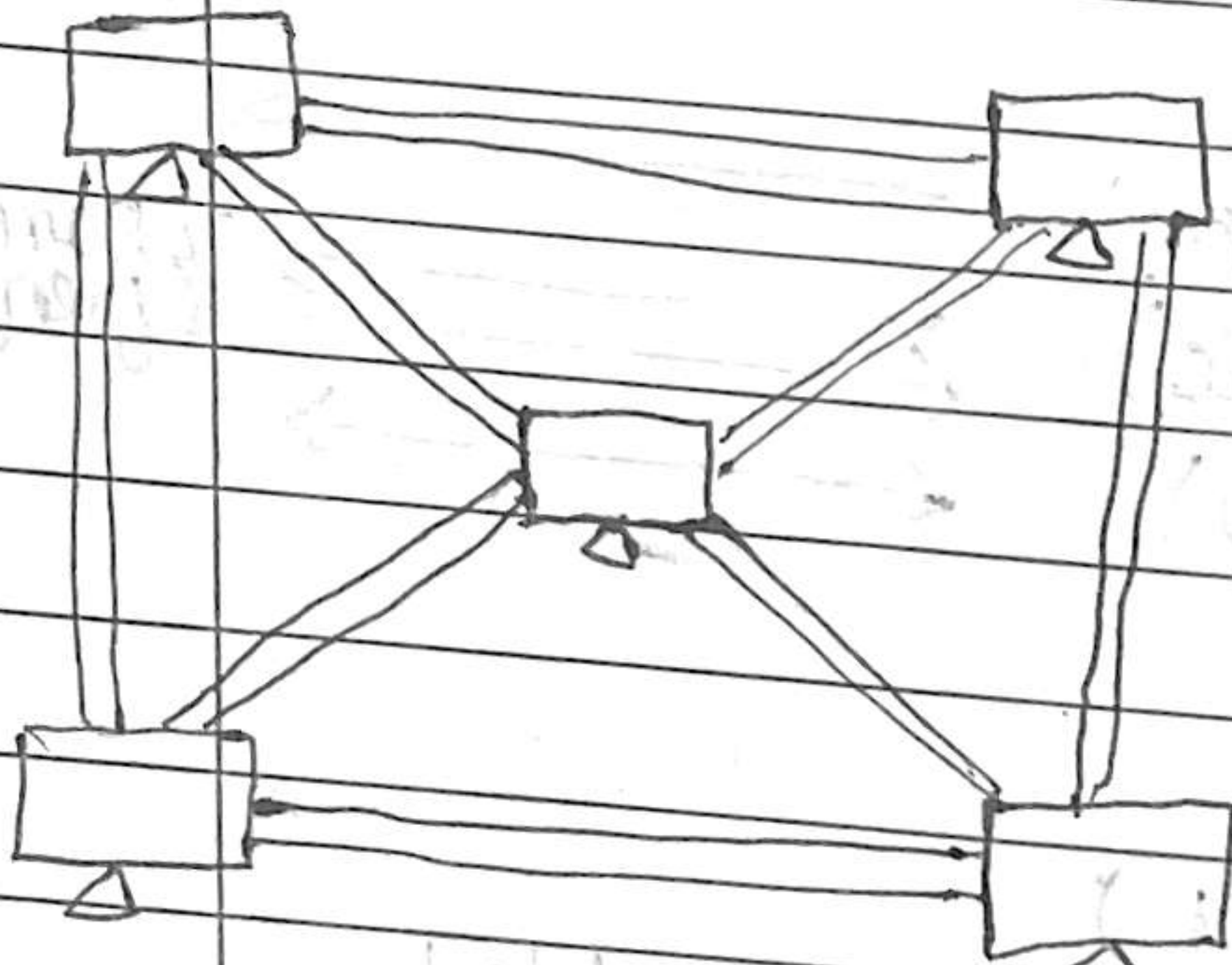
CSMA/CD
CSMA/CA



RING
TOPOLOGY



TREE
TOPOLOGY



MESH
TOPOLOGY

Hybrid Topology:
Mix of more than 1
topology EXCEPT mesh

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OSI (Open System Interconnected)
(OR)

ISO (International Standardized Organization)

- Developed by European Telecom Industry in mid 1960s
- To standardize network communication

- Framework consists of 7 layers

(i) Application (First layer, provides interface between human & machine)

(ii) Presentation

(iii) Session

(iv) Transport (Heart of OSI)

(v) Network

LLC

(vi) Data Link

MAC

(vii) Physical

Please do not touch
steel pillar ~~appliance~~
appliance

Presentation ^{→ sender}

[do encryption { convert plain / write text }
[decryption { to cipher text }

→ receiver

[encoding (sender)

[decoding (receiver)

[compression (sender)

[decompression (receiver)

MNEMONIC for OSI model

All → Application

People → Presentation

seem → Session

To → Transport

Need → Network

Data → Data Link

Processing → Physical

Session

- Responsible for 3 things:

(i) Connection establishment

(ii) Data transmission (monitoring)

(iii) Connection Termination

Transport

- Transmitting information from one place to another

Network (Internet Layer)

- Deals with IP (Internet Protocol) Address / logical address
- For packet numbering (rearranges the packets)

Data Link

- (i) LLC (Logic Linked Control) $\Rightarrow$ error detection
- (ii) MAC (Medium Access Control) $\Rightarrow$ deals with physical address

Physical

- Deals with transferring the data in 0 and 1 form

* Mbps bits
 MBps bytes

Application Presentation Session } Original data ; high level layers / application layers / top layers

Transport $\rightarrow$ Heart of OSI (Segment $\Rightarrow$ small blocks)

Host Layers / Bottom Layer

Network $\rightarrow$ divided into packets (easier to do packet numbering)
 Data Link $\rightarrow$ frames
 Physical $\rightarrow$ bits } Link Layers

Host Layers = Internet Layer + Link Layers

Developed by ARPANET

Protocol

TCP/IP has 4 layers

- Application layer
- Transport layer
- Internet layer
- Link Layer

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Developed in ARPANET in 1981

T
T
N

192.168.10.12
octals

~~HTTP~~ HTTPS

~~HTTP~~ Protocols -

ping www.google.com
↳ DNS Register

A 1) HTTPS } Websites Websites

A 2) HTTP }

A 3) SNMP → To manage the network & use faster route

A 4) POP3 → Receive emails from mail server

A 5) TELNET → session establishment & session termination.

R 6) FTP → Uploading & downloading of files

A 7) SMTP → ~~For emails~~ For emails

A 8) TFTP → Faster version of FTP

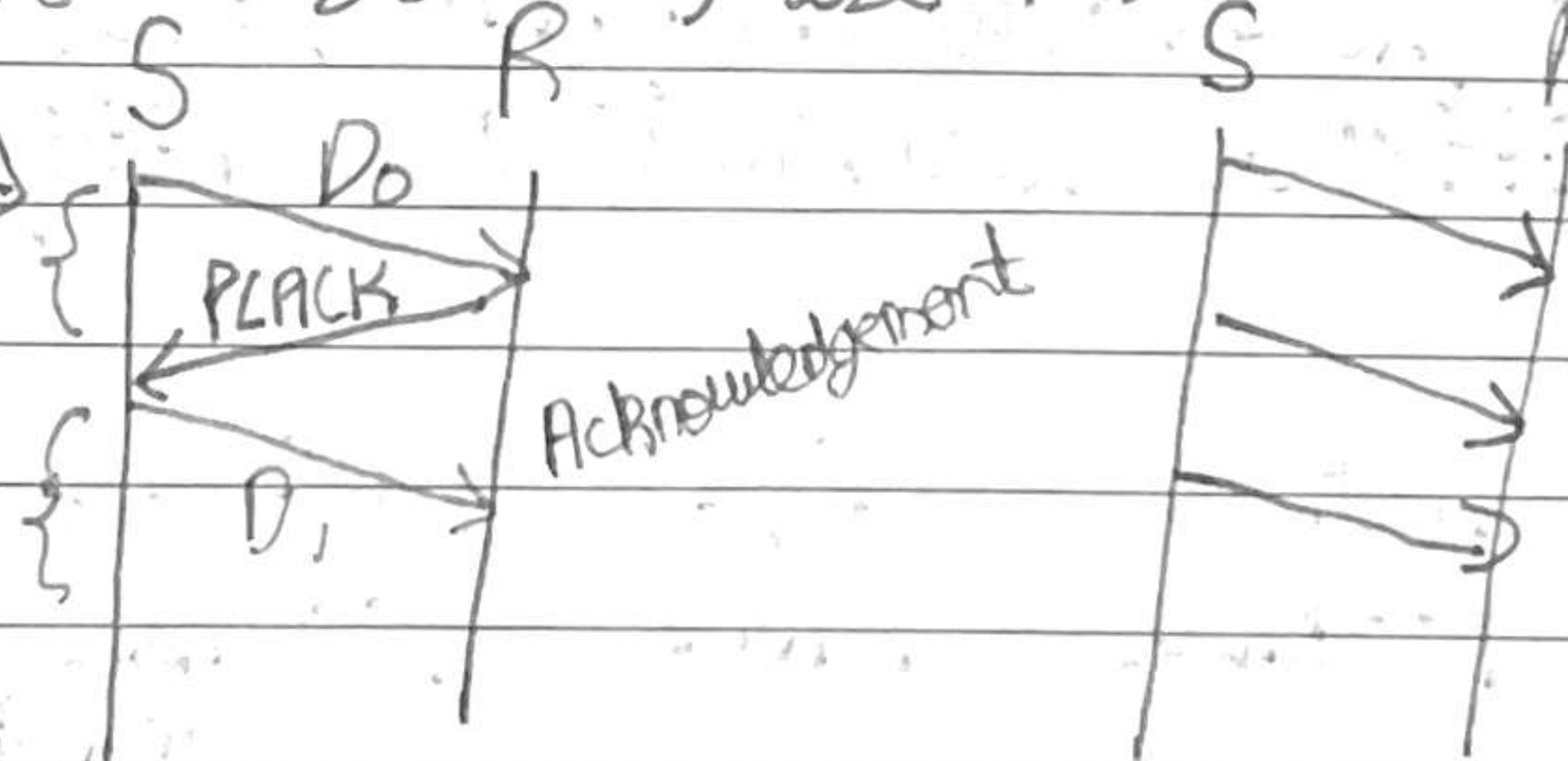
9) TCP → wired, reliable, slower, variable header length

10) UDP → wireless, unreliable, faster, fixed header length

11) IP →

↳ IPv4 - 4 octal ; use numbers

↳ IPv6 - 8 octal ; use hexadecimal (1-9 A-F)



N 12) ARP → using IP, map IP to MAC

Map MAC
↑ to IP

N 13) RARP → using MAC, we can find IP address

N 14) ~~ICMP~~ → send or receive echo message (linked with SNMP)

PD 15) IEEE802.3 → ethernet

D 16) PPP → Direct point to point data link communication

D 17) CSMA/CD → Avoid collisions in Ethernet

P 18) SONET → High speed optical fiber transmission

T 19) SCTP → media streaming

CISCO Packet Receiver

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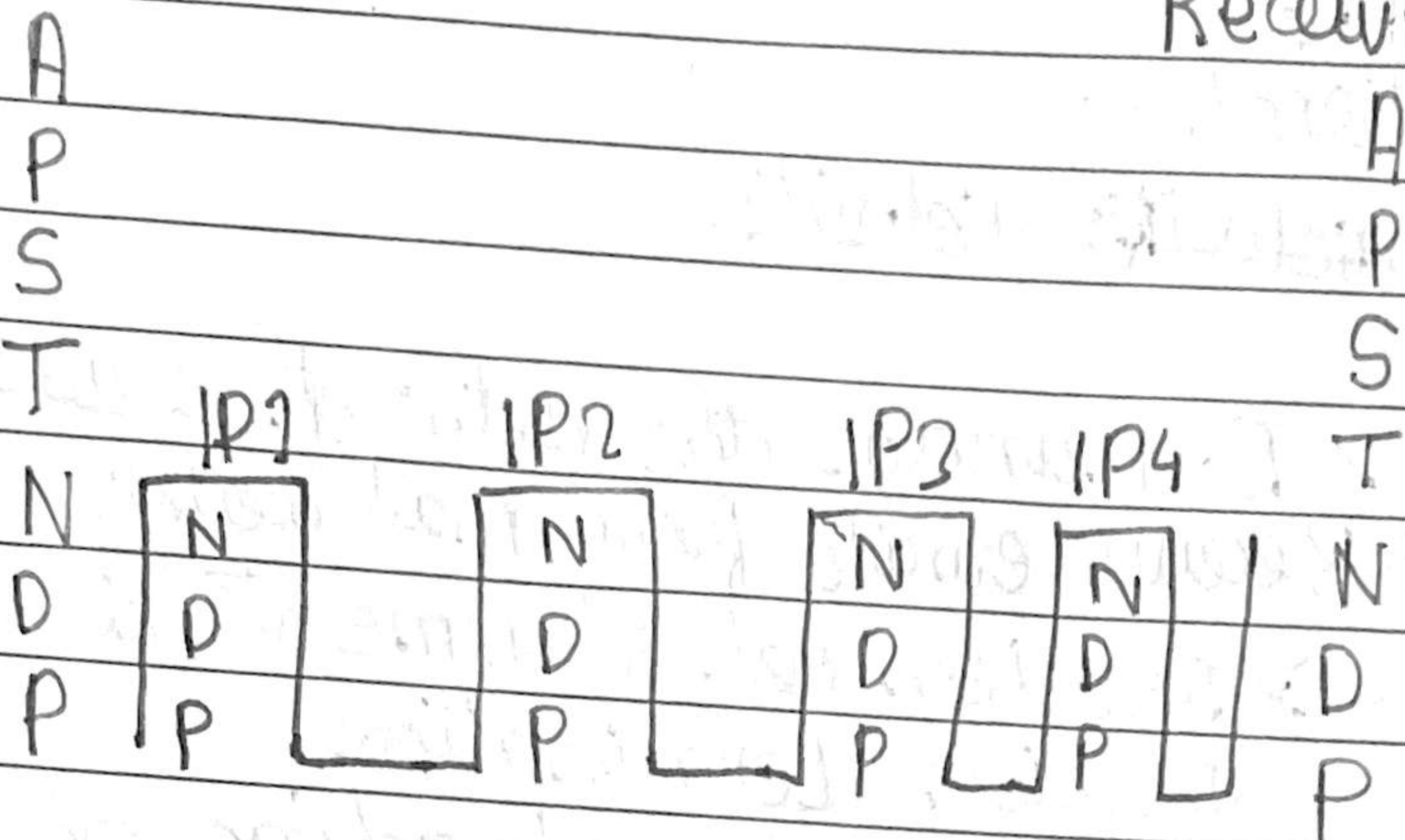
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Sender

Receiver / IP5



a single stop a packet takes from one device to next

HOP → Connecting points

Intermediate Node

Adjacent Node

Neighbouring Node

Ethernet ⇒ IEEE 802.3

Coaxial ⇒ IEEE 802.4

Twisted ⇒ IEEE 802.5

WLAN

Ethernet (Xerox Palo Alto 1976) / PIC / NIC card
⇒ IEEE 802.3 (Standardized by IEEE, 1983)

Standard
10Mbps

fast
100Mbps

Giga
1Gbps

fast Giga
100Gbps

Pay load

(OR)

User Data

(OR)

Data

Ethernet

→ Coaxial ⇒ Token Bus ⇒ IEEE 802.4

→ Twisted ⇒ Token Ring ⇒ IEEE 802.5

STP

8 wires

RJ-45

clockwise; physical

Topology

anti clockwise; virtual

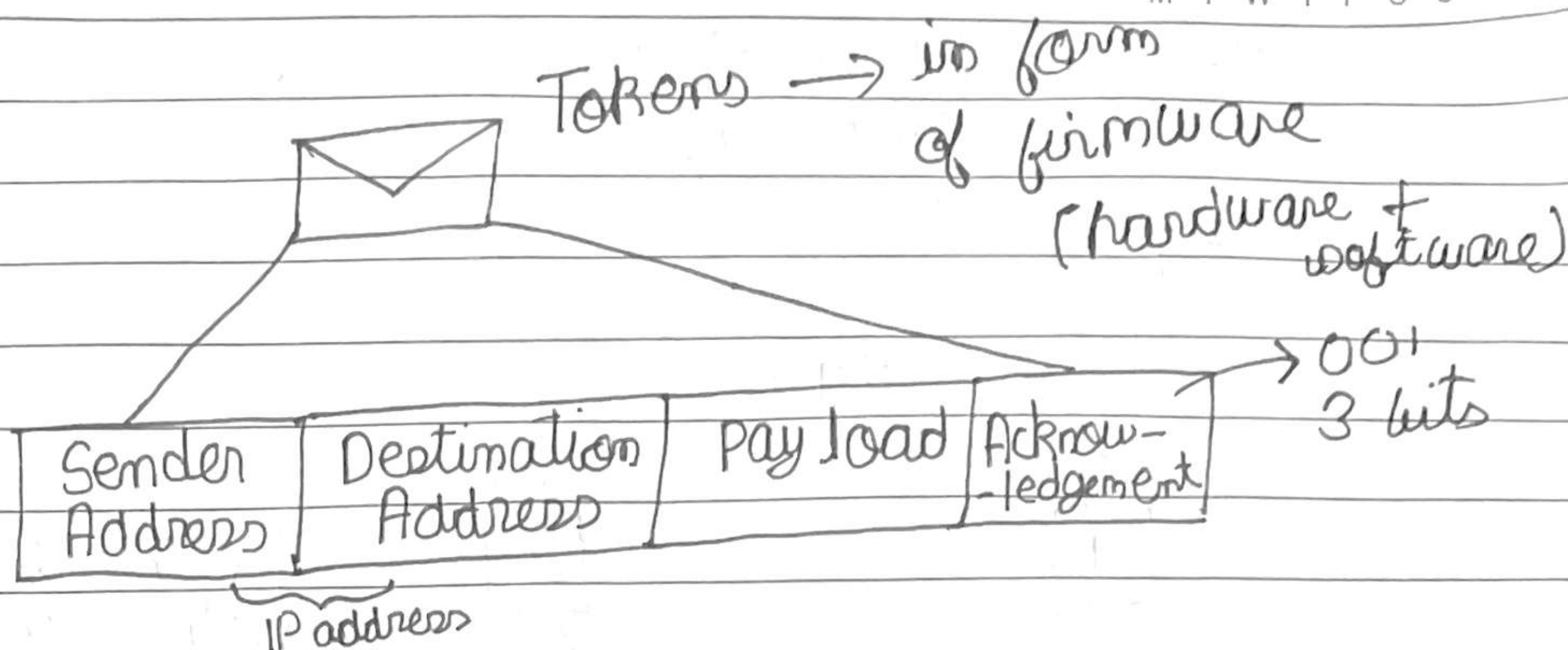
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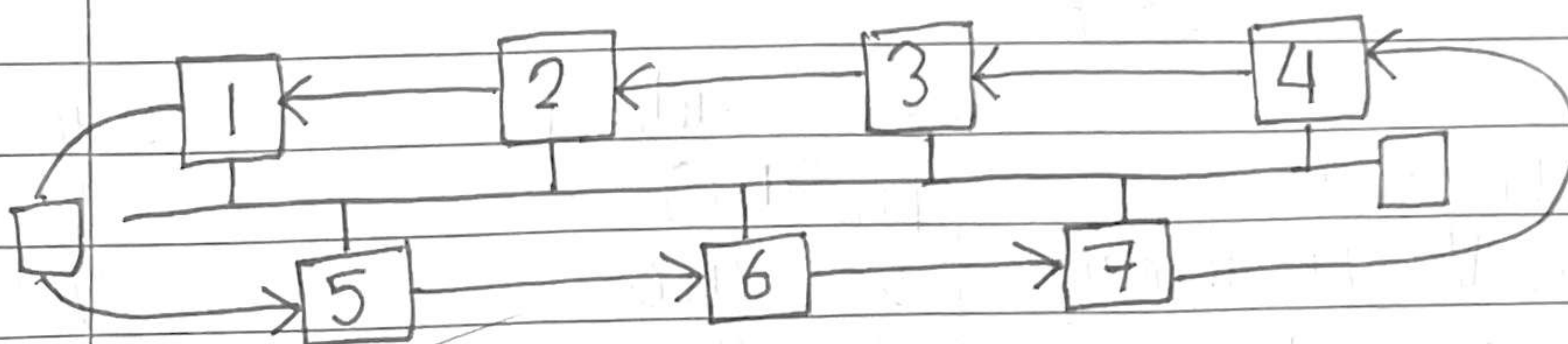
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TOKEN BUS $\Rightarrow$ IEEE 802.4



$\Rightarrow$ Turn Around Time = Waiting time + Processing time

ASSIGNMENTS -

(With CU Cover Page)

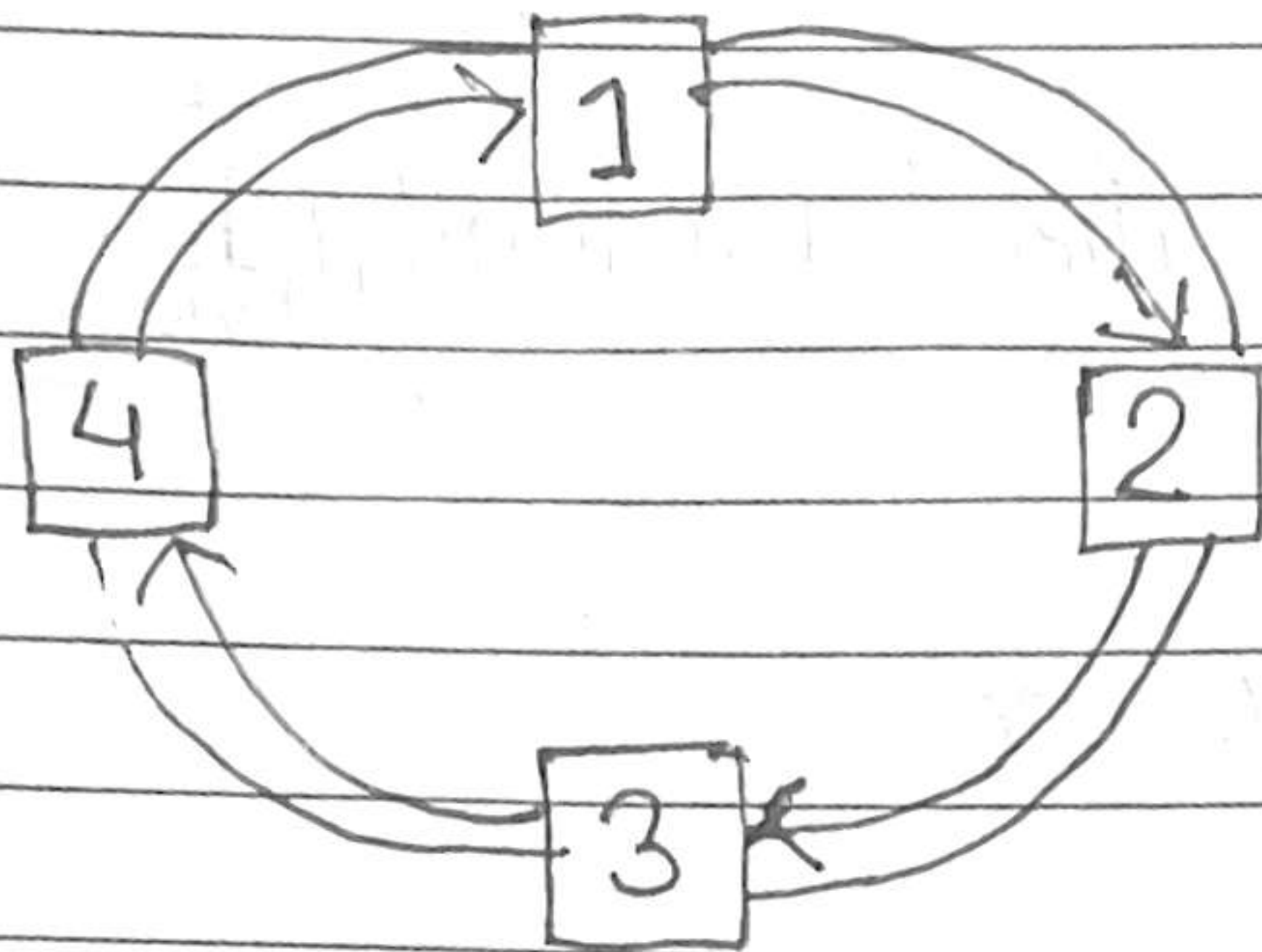
Deadline: 14 August 2025 (Before Noon)

- 1) Write short note on mode of communications (data flow), medium (with pictures), types of crimping, and different types of topologies (with figures)
- 2) Explain in detail about OSI layers.

TOKEN RING $\Rightarrow$ IEEE 802.5

Shielded Twisted Pair

8 pins (RJ-45)



To connect different nature of devices

Crimping-

(i) Straight through

(ii) Cross over

Side A

Side B

Pin 1	O	→	O
2	W/O	→	W/O
3	B	→	B
4	W/B	→	W/B
5	G	→	G
6	W/G	→	W/G
7	Bn	→	Bn
8	W/Bn	→	W/Bn

Side A

Side B

Go to Cisco

use college ID

& install

Packet Tracer

Pin 1	O	→	W/O Pin 1
2	W/O	→	W/O → W/G 2
3	B	→	O 3
4	W/B	→	W/B → W/B 4
5	G	→	G → G 5
6	W/G	→	W/G → B 6
7	Bn	→	Bn → Bn 7
8	W/Bn	→	W/Bn → W/Bn 8

To connect same nature of devices

hub ↔ hub

switch ↔ switch

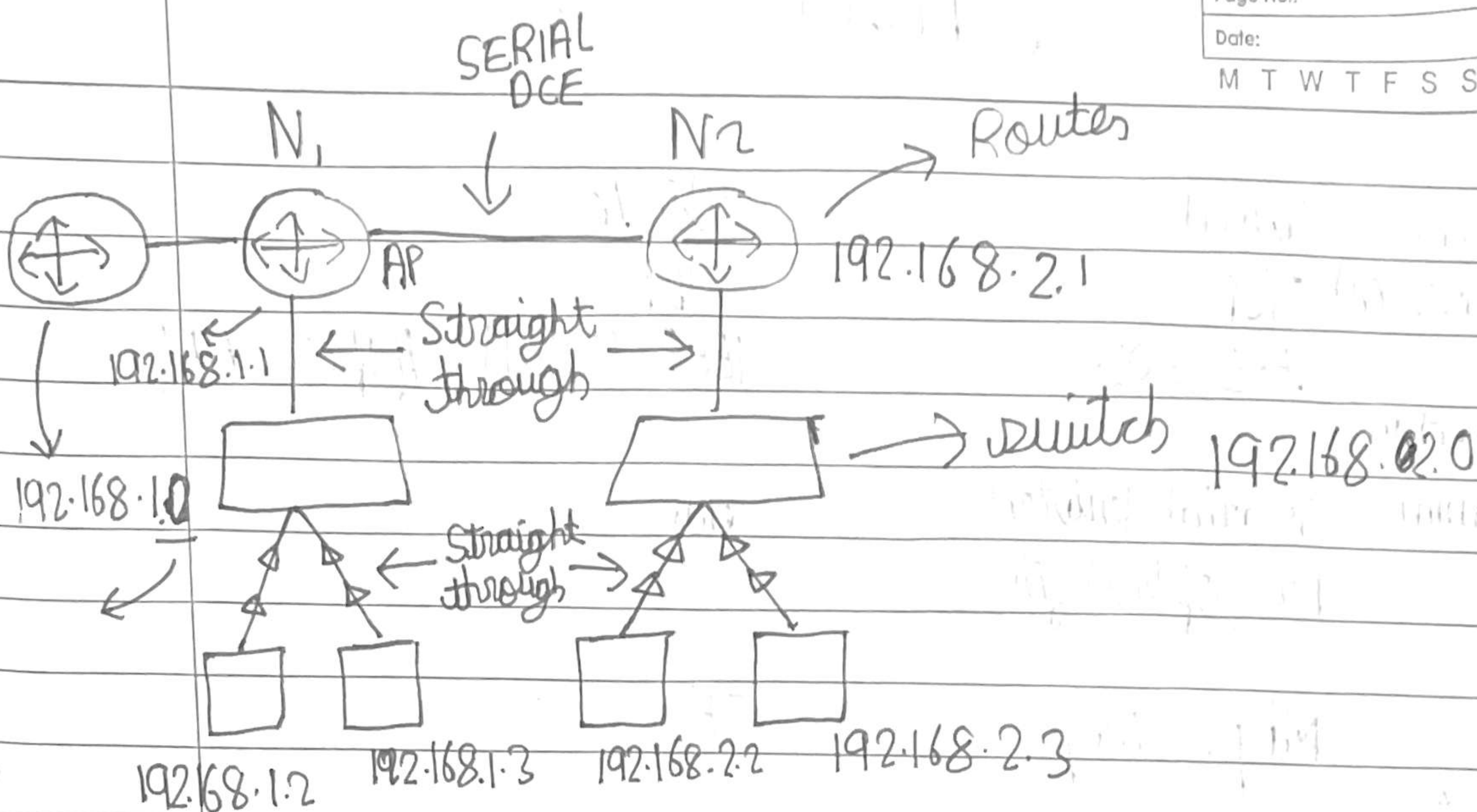
AP = Access Point

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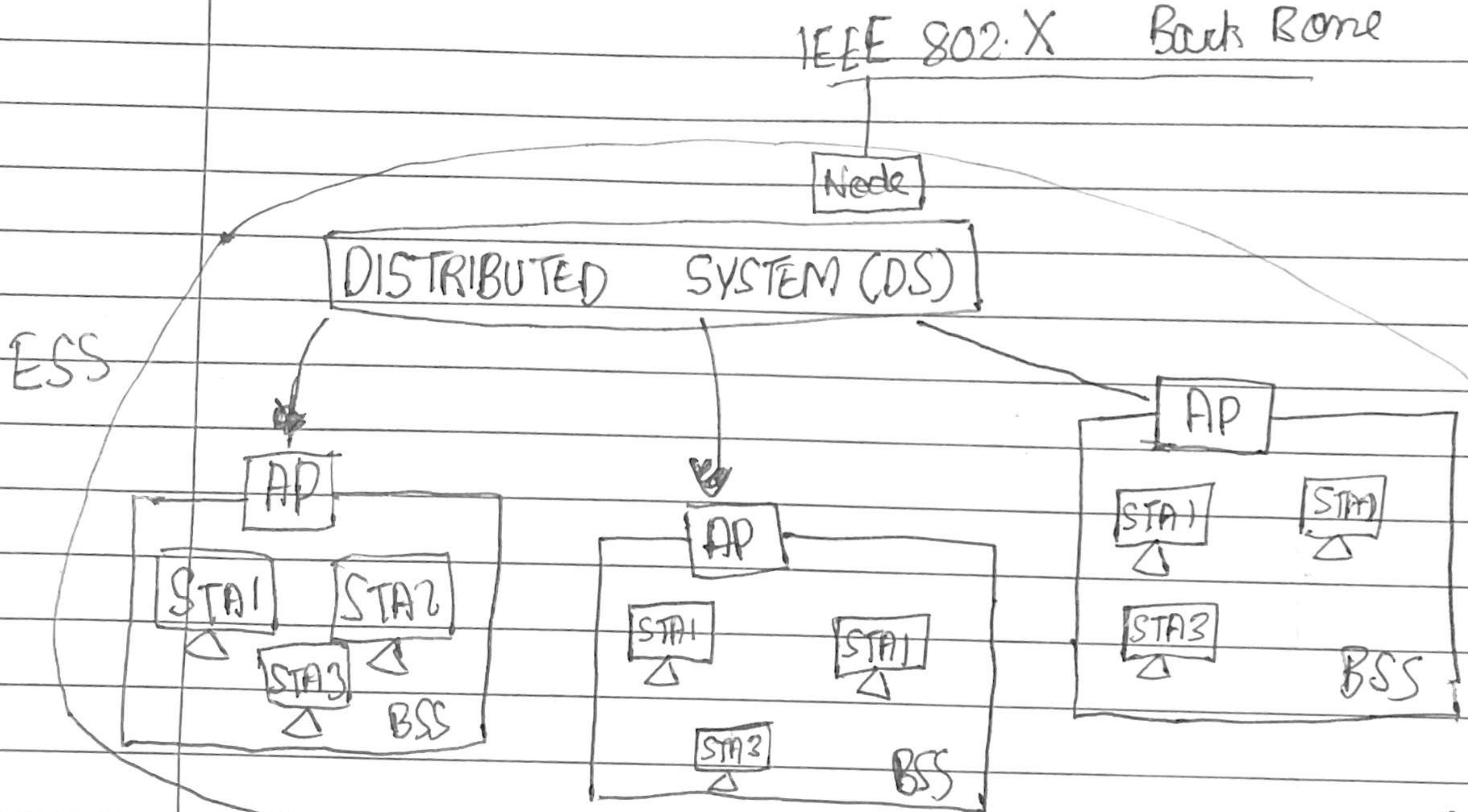
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Aspect	Wired	Wireless
1) Protocol	TCP	UDP
2) IEEE standard	IEEE 802.3	IEEE 802.11 (Hyper LAN)
3) Medium	Coaxial, Twisted Pairs, Optical fiber	Air
4) Data Transfer Rate	Mbps / MBps	GHz
5) Data format while transfer	Electrical (or) light signals	Radio frequency



Types -

(i) Infrastructure

(ii) Ad-hoc

MANET

VANET

↳ hotspot

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Wireless

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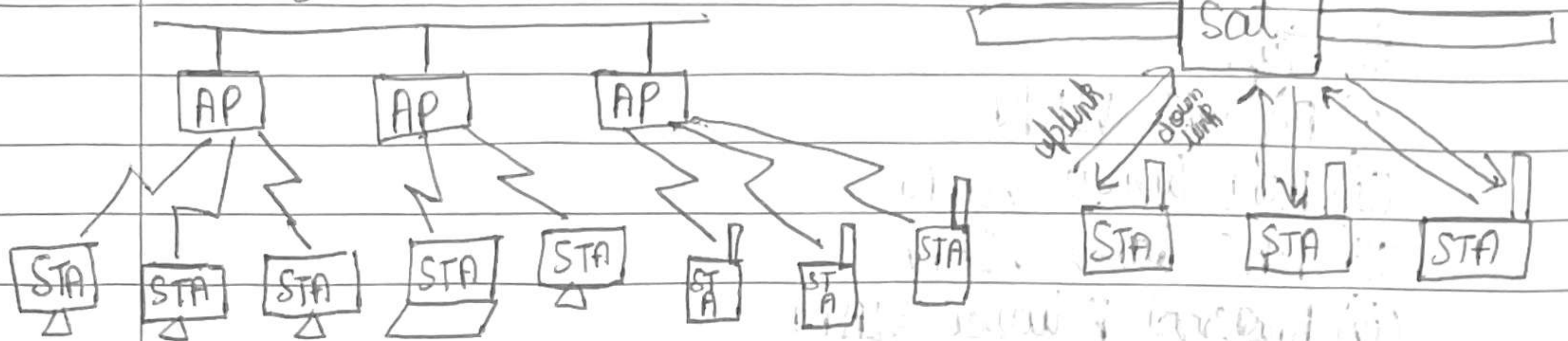
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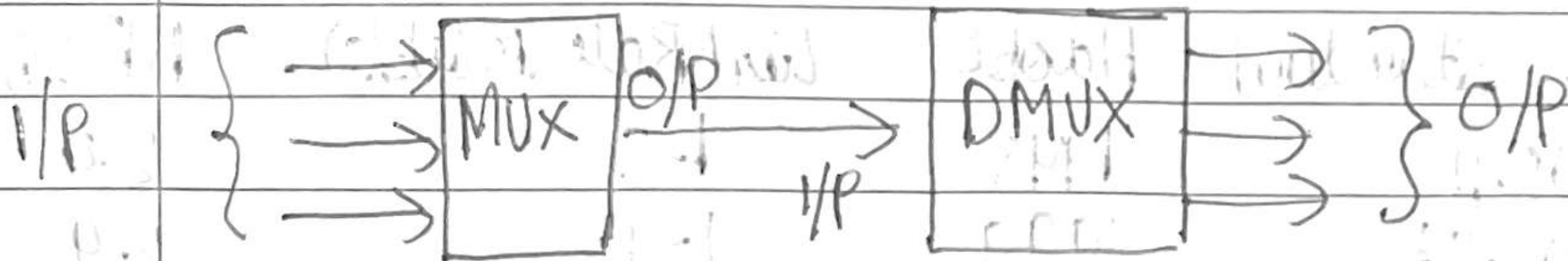
Types of ~~WIRELESS~~ LAN -
- Infrastructure

- Adhoc

IEEE 802.3 Backbone



Multiplexing -



- Space Division Multiple Access (SDMA)

- Frequency Div. Multiple Access (FDMA)

- Time Division Multiple Access (TDMA)

- Code Division Multiple Access (CDMA)

- Orthogonal Frequency Division Multiple

info stored in receiver
buffer after transmission
can be decoded

info stored after transmission
oscillator needs to
streamline the freq

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VLAN -

Virtual Local Area Network
For isolation of Users

Types

(i) Default VLAN

(ii) Port based VLAN

(iii) MAC based VLAN

(iv) Protocol based VLAN

MIMO
concept

IEEE 802.11 Types -

Gen	IEEE Standard	Adopt	Link Rate (Mbps)	RF (GHz)
Wi-Fi 1	802.11	1997	1-2	2.4
Wi-Fi 2	802.11b	1999	1-11	2.4
Wi-Fi 3	802.11a	1999	6-54	5
OFDM →	802.11g	2003	6-54	2.4
Wi-Fi 4	802.11n	2009	6.5-600	2.4, 5
Wi-Fi 5	802.11ac	2013	6.5-6983	5(a)
Wi-Fi 6	802.11ax	2021	0.4-9608	2.4, 5
Wi-Fi 6E			0.4-23,059	2.4, 5, 6
Wi-Fi 7	802.11be	2024	0.4-23,059	2.4, 5, 6
Wi-Fi 8	802.11bn		100,000 [4]	2.4, 5, 6

under development

don't

harm

living

beings

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Propagation Delay -

- Time taken by a signal to travel from sender to receiver through a transmission medium
- It depends on the distance & the propagation speed of the signal in the medium.

Stop and Wait Protocol -

- In Stop-and-Wait ARQ, the sender transmits one frame and then waits until an acknowledgment (ACK) is received before sending the next frame.
- Channel alternates between sending data & waiting for ACKs.
- Makes protocol simple but inefficient especially if product of bandwidth $\times$ delay (bandwidth delay product) is high, because much of the link capacity remains idle.

Round Trip Time (RTT) -

- Time taken for a signal to travel from the sender to receiver and back (including propagation + processing + queuing delays).
- In Stop & Wait RTT determines how much time the sender waits after sending a frame until it can send you.

Q. A link has a bandwidth of 2 Mbps & RTT is 30 ms. Each frame is 500 bytes. What will be the efficiency of the Stop & Wait protocol.

Ans Bandwidth (B) = 2 Mbps = 2×10^6 bits/sec
RTT = 30 ms = 0.03 s

Frame size = 500 bytes = $500 \times 8 = 4000$ bits

Transmission Time per frame —

$$T_{\text{frame}} = \text{frame size} / \text{Bandwidth}$$

$$T_{\text{frame}} = 4000 / (2 \times 10^6) = 0.002 \text{ s} = 2 \text{ ms}$$

Cycle Time —

$$T_{\text{cycle}} = T_{\text{frame}} + \text{RTT}$$

$$T_{\text{cycle}} = 0.002 + 0.03 = 0.032 \text{ s}$$

Efficiency of Stop & Wait —

$$\eta = \text{Transmission time} / \text{Cycle Time}$$

$$\eta = 0.002 / 0.032 = 0.0625$$

$$\text{Efficiency } (\eta) = 6.25\%$$

Efficiency of Stop & Wait protocol is very low. It is only 6.25% in this case. This is because the time taken for the frame to be transmitted is very small compared to the time taken for the frame to be received and for the receiver to send back the acknowledgment. This is why the Stop & Wait protocol is not used in high-speed networks.

RTT (Round Trip Time) is the time taken for a frame to be transmitted from the sender to the receiver and for the receiver to send back the acknowledgment. It is the sum of the transmission time and the propagation time. In this case, the transmission time is 0.002 s and the propagation time is 0.03 s. Therefore, the RTT is 0.032 s.

4. TTA (Total Time for Acknowledgment) is the time taken for the frame to be transmitted from the sender to the receiver and for the receiver to send back the acknowledgment. It is the sum of the transmission time and the propagation time. In this case, the transmission time is 0.002 s and the propagation time is 0.03 s. Therefore, the TTA is 0.032 s.